# Conditions for eigenvalue configurations of two real symmetric matrices: a stochastic definition

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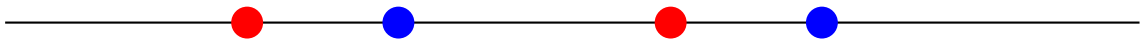
# Problem

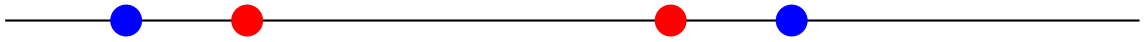
**Definition (Eigenvalue Configuration)**

Let $F \in \mathbb{R}^{m\times m}$, $G \in \mathbb{R}^{n\times n}$ be symmetric.
Then their **eigenvalue configuration**, which we write as $\mathrm{EC}(F, G)$, is the "relative locations" of the eigenvalues.

Example 1: Let $F = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}$ and $G = \begin{bmatrix} 2 & -1 \\ -1 & 4 \end{bmatrix}$.

$EC(F, G) =$  $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Example 2: Let $F = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}$ and $G = \begin{bmatrix} 0 & -1 \\ -1 & 4 \end{bmatrix}$

$EC(F, G) =$  $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

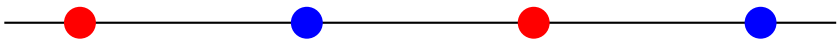
# Problem

## Problem

*In* : $\mathbf{c}$, a desired eigenvalue configuration

*Out*: Condition on $F = [a_{ij}]$ and $G = [b_{ij}]$ such that $\mathrm{EC}(F, G) = \mathbf{c}$

Example:

*In* : $\mathbf{c} =$  $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

*Out*: $\begin{pmatrix} \sigma(D_{00}) \\ \sigma(D_{01}) \\ \sigma(D_{10}) \\ \sigma(D_{11}) \end{pmatrix} \in \left\{ \begin{pmatrix} 1 \\ 2 \\ -1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \\ 0 \\ 0 \end{pmatrix} \right\}$ where

$$D_{00} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$D_{01} = \begin{bmatrix} -a_{2,2} - a_{1,1} + 2b_{1,1} & 2b_{1,2} \\ 2b_{1,2} & -a_{1,1} - a_{1,1} + 2b_{2,2} \end{bmatrix}$$

$$D_{11} = \begin{bmatrix} a_{1,1}a_{2,2} - {a_{1,2}}^2 - (a_{1,1} + a_{2,2})\, b_{1,1} + {b_{1,1}}^2 + {b_{1,2}}^2 & -b_{1,2}\,(a_{1,1} + a_{2,2} - b_{1,1} - b_{2,2}) \\ -b_{1,2}\,(a_{1,1} + a_{2,2} - b_{1,1} - b_{2,2}) & a_{1,1}a_{2,2} - {a_{1,2}}^2 - (a_{1,1} + a_{2,2})\, b_{2,2} + {b_{1,2}}^2 + {b_{2,2}}^2 \end{bmatrix}$$

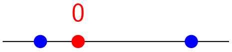
$$\vdots$$

# Problem: Background and Motivation

Generalization of Descartes' rule of signs: ([1, 2])

$$F = [0] \quad \in \mathbb{R}^{1\times 1} \qquad \text{and} \qquad G = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \in \mathbb{R}^{2\times 2}$$

$g(x) = \det(xI - G) = (x+1)(x-3)$




Descartes' rule of signs:

$$\begin{aligned} g(x) = \underbrace{x^2}_{+}\underbrace{-2x}_{-}\underbrace{-3}_{-} & \\ \implies v(+,-,-) &= 1 \\ &= \#\ \text{positive roots of } g \\ &= \mathrm{EC}(F, G). \end{aligned}$$

# Problem: Background and Motivation

Applications in Science and Engineering: Many nontrivial problems in science/engineering can be reduced to this problem.

Example (network analysis [3]):

- Let $F$ be the adjacency matrix of a graph.
- Some properties are related to the eigenvalues of $F$.
- How do eigenvalues change when edges are added or removed?
- Changing edges corresponds to adding a matrix $U$ to $F$
- Hence, one might be interested in the eigenvalue configuration of

$$F \qquad \text{and} \qquad G = F + U.$$

# Problem

## Non-triviality

Why is this problem nontrivial?

Nontrivial specialized quantifier elimination problem.

**Problem (Rephrased)**

*In* : $\underset{\alpha_1,\ldots,\alpha_m,\ \beta_1,\ldots,\beta_n}{\exists} \Phi_{\mathbf{c}}(\alpha_1,\ldots,\alpha_m,\beta_1,\ldots,\beta_n,a_{ij},b_{ij})$

*Out*: $\Psi_{\mathbf{c}}(a_{ij},b_{ij})$ equivalent to the above.

- Tarski [4] gave first algorithm for quantifier elimination of general formulas over real closed fields.
- Improvements: Collins, Hong, McCallum, Grigorev, Roy, Renegar, Canny, Brown, Strzebonski, Safey-Eldin, ..., Chen [5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21]

Q: Why not just use existing algorithms for quantifier elimination?

A: General algorithms not good for specialized input.

**Challenge:** Develop an algorithm that exploits the problem structure.

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# Generic Case: Setup

- Let $F = [a_{ij}] \in \mathbb{R}^{m\times m}$ and $G = [b_{ij}] \in \mathbb{R}^{n\times n}$ be symmetric.
- For now: let $F$ and $G$ be **generic**; i.e. they don't share eigenvalues.
- Label the eigenvalues $\alpha_1 \leq \cdots \leq \alpha_m$ and $\beta_1 \leq \cdots \leq \beta_n$, respectively.
- Let $f$ denote the characteristic polynomial of $F$.

# Generic Case: Definition

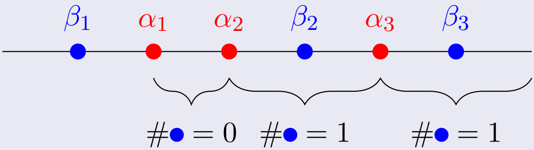
## Definition (Generic)

Let $F \in \mathbb{R}^{m\times m}$, $G \in \mathbb{R}^{n\times n}$ be symmetric and generic (i.e. having **distinct eigenvalues**).

Then the **generic** eigenvalue configuration $\mathrm{GEC}(F,G)$ is the vector $\mathbf{c} \in \mathbb{N}^{m\times m}$ where

$$\mathbf{c}_t = \#\{j : \beta_j \in (\alpha_t, \alpha_{t+1})\}.$$

## Example




$$\mathrm{GEC}(F,G) = (0,1,1).$$

# Generic Case: Theorem Statement

## Theorem

We have $\mathbf{c} = \mathrm{GEC}(F, G) \iff \mathbf{c} = C_{\text{sig}}\ A_{\text{sig}}(F, G)$, where

1. $C_{\text{sig}} = VH^{-1} \in \mathbb{N}^{m \times 2^m}$, where

$$V_{t,s} = \delta_{\{v(s,+), m-t\}} \qquad t \in \{1, \ldots, m\},\ s \in \{-,+\}^m$$
$$H_{e,s} = s_1^{e_1} \cdots s_m{}^{e_m} \qquad e \in \{0,1\}^m, \quad s \in \{-,+\}^m$$

2. $A_{\text{sig}}(F, G) \in \mathbb{Z}^{2^m}$, where

$$(A_{\text{sig}})_e = \sigma\left(f_e(G)\right) \qquad e \in \{0,1\}^m$$
$$f_e = f^{(0)^{e_0}} \cdots f^{(m-1)^{e_{m-1}}} \qquad f^{(k)} = k\text{-th derivative}$$
$$f = \det(xI_m - F).$$

Remark: Since all the eigenvalues are real, one can easily express the signature in terms of sign variation count and the entries of $F$ and $G$.

# Generic Case: Proof Outline

Sketch of Proof: (with full details: ≈24 pages in [22])

$$\mathbf{c} = \mathrm{GEC}(F, G)$$

$$\Updownarrow$$

$$\mathbf{c}_t = \#\{j : \alpha_t < \beta_j < \alpha_{t+1}\}$$

$$\Updownarrow$$

$$\mathbf{c}_t = \#\{j : \overline{\#}\{x : f(x) = 0 \wedge x > \beta_j\} = m - t\}$$

$$\Updownarrow$$

$$\mathbf{c} = V \begin{bmatrix} \#\{j : \mathrm{sign}\left(f^{(0)}(\beta_j), \ldots, f^{(m-1)}(\beta_j)\right) = - \cdots -\} \\ \vdots \end{bmatrix}$$

$$\Updownarrow$$

$$\mathbf{c} = VH^{-1} \begin{bmatrix} \sigma\left(\left(f^{(0)}\right)^0 \cdots \left(f^{(m-1)}\right)^0 (G)\right) \\ \vdots \end{bmatrix}$$

$$\Updownarrow$$

$$\mathbf{c} = C_{\mathrm{sig}}\ A_{\mathrm{sig}}(F, G).$$

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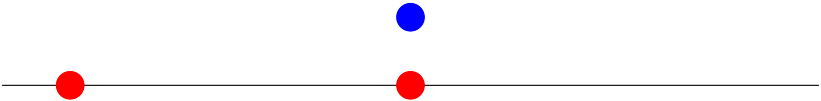
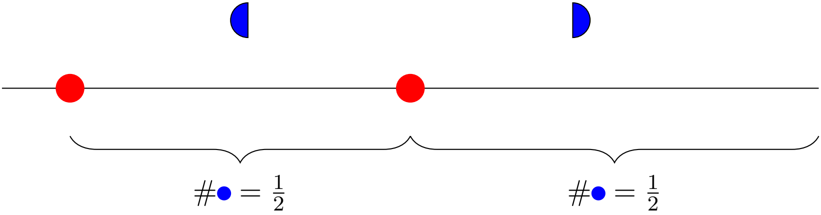
# Shared Eigenvalues: Idea



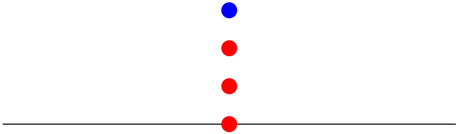
Q: What do we get when we run the algorithm?

A:

$$\mathrm{EC}(F, G) = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$$



# Shared Eigenvalues: Idea



$$\mathrm{EC}(\color{red}{F}, \color{blue}{G}) = \begin{bmatrix} 3/8 \\ 3/8 \\ 1/8 \end{bmatrix}$$

# Shared Eigenvalues: New Definition

## Definition (Generalized Eigenvalue Configuration)

Let $F$ and $G$ be arbitrary real symmetric matrices. Then

$$\begin{aligned}\mathrm{EC}(F,G) &= \mathbb{E}\ \mathrm{GEC}\left(F_d, G\right) \\ &= \frac{1}{2^m} \sum_{d\in\{-\varepsilon,\varepsilon\}^m} \mathrm{GEC}\left(F_d, G\right)\end{aligned}$$

where

$$\begin{aligned} d &\in \{-\varepsilon,\varepsilon\}^m \text{ i.i.d. random variables} \\ F_d &= \text{matrix with eigenvalues } \alpha_i + d_i \\ \varepsilon &= \text{"small enough" positive number.}\end{aligned}$$

# Shared Eigenvalues: New Definition

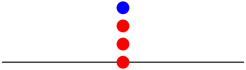
## Definition

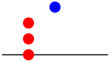
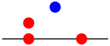
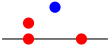
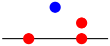
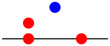
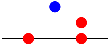
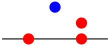
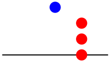
$$\begin{aligned}\mathrm{EC}(F, G) &= \mathbb{E}\ \mathrm{GEC}\left(F_d, G\right) \\ &= \frac{1}{2^m} \sum_{d \in \{-\varepsilon, \varepsilon\}^m} \mathrm{GEC}\left(F_d, G\right)\end{aligned}$$

Some remarks:

- $\varepsilon$ chosen so that $F_d$ and $G$ do not share eigenvalues.
- $\varepsilon$ chosen so that perturbed red points do not move too far.
- New definition still works when $F$ and $G$ do not share eigenvalues.

# Shared Eigenvalues: Example



| $d$ | $\mathbf{c}$ | EC | $d$ | $\mathbf{c}$ | EC |
|---|---|---|---|---|---|
| $- - -$ |  | $\begin{bmatrix}0\\0\\1\end{bmatrix}$ | $+ - -$ |  | $\begin{bmatrix}0\\1\\0\end{bmatrix}$ |
| $- - +$ |  | $\begin{bmatrix}0\\1\\0\end{bmatrix}$ | $+ - +$ |  | $\begin{bmatrix}1\\0\\0\end{bmatrix}$ |
| $- + -$ |  | $\begin{bmatrix}0\\1\\0\end{bmatrix}$ | $+ + -$ |  | $\begin{bmatrix}1\\0\\0\end{bmatrix}$ |
| $- + +$ |  | $\begin{bmatrix}1\\0\\0\end{bmatrix}$ | $+ + +$ |  | $\begin{bmatrix}0\\0\\0\end{bmatrix}$ |

$$\mathrm{EC}({\color{red}F},{\color{blue}G}) = \frac{1}{8}\left(\begin{bmatrix}0\\0\\1\end{bmatrix} + 3\begin{bmatrix}0\\1\\0\end{bmatrix} + 3\begin{bmatrix}1\\0\\0\end{bmatrix} + \begin{bmatrix}0\\0\\0\end{bmatrix}\right) = \begin{bmatrix}3/8\\3/8\\1/8\end{bmatrix}$$

# Shared Eigenvalues: Proof Outline

Sketch of Proof: (with full details: ≈11 pages in [22])

$$\begin{tikzcd}
\mathrm{EC}(F,G) \arrow[rr, equal, "?"] \arrow[d, equal, "\text{decompose}"'] & & C_{\mathrm{sig}}\ A_{\mathrm{sig}}(F,G) \arrow[d, equal, "\text{combine}"] \\
\sum_{j=1}^{n} \mathrm{EC}(F,[\beta_j]) \arrow[rr, equal] & & \sum_{j=1}^{n} C_{\mathrm{sig}}\ A_{\mathrm{sig}}(F,[\beta_j])
\end{tikzcd}$$

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# Future Work

Short-term:

- Find applications
- Prune redundancies from output

Long-term:

- Generalize to $> 2$ matrices
- Generalize to tensors

Other solutions:

- We have another algorithm for $\mathrm{GEC}$ which uses the theory of symmetric polynomials [23].
- Can we find more?

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*Thank you!*